

## DATASET REPORT

### A 15-Year Multivariate Hydrometeorological and Water Level Dataset with 7-Day Lags along the Ganges–Padma River Corridor (2011–2025)

|                  |                                                                                                                                                                          |                        |                                                                                                                                                         |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Group No.</b> | Group 01                                                                                                                                                                 | <b>Dataset Version</b> | v2.0 (September 2026)                                                                                                                                   |
| <b>Students</b>  | Md. Shakhoyat Rahman Shujon (2107104)<br>Mohammad Moin Uddin Moin (2107111)<br>Md. Tariful Islam Jony (2107119)<br>Kamrul Islam (2107102)<br>Tanha Sayed Ahona (2107093) | <b>Repository</b>      | <a href="https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset">https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset</a> |

(a) Framework illustrating the practical application of the data

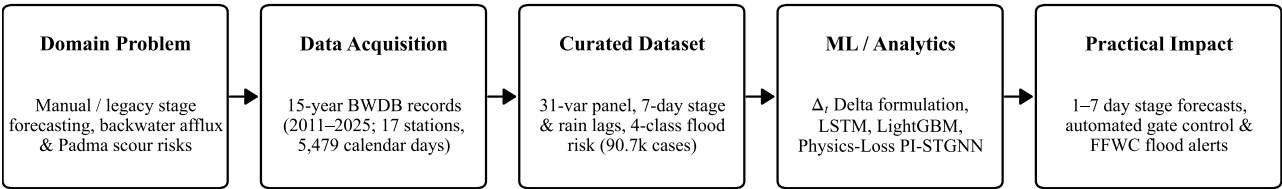


Figure 1(a). Dataset-specific practical application framework illustrating end-to-end impact from hydrometric collection to automated river corridor warning and bridge operations.

(b) Technical pipeline of the ML method using the dataset

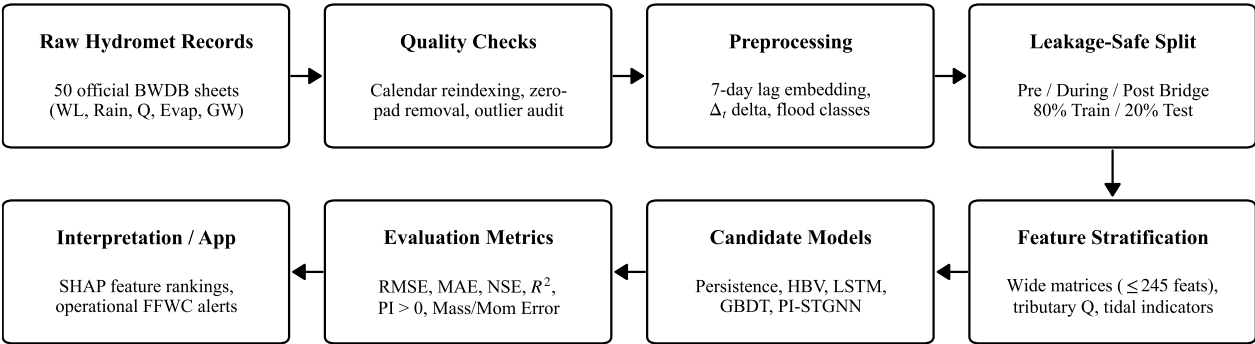


Figure 1(b). Exact technical pipeline showing leakage-safe partitioning, multivariate lag matrix generation, physics-guided architectures, and persistence-benchmarked evaluation.

# 1 Dataset Article Information

**Table 1:** Dataset Article Metadata and Administration.

| Item                              | Description / Metadata                                                                                                                                                                                                                                                           |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dataset/report title</b>       | Multi-Variable Hydrometeorological and Point-Wise Water Level Dataset with 7-Day Lags along the Ganges–Padma River Corridor (2011–2025)                                                                                                                                          |
| <b>Group and students</b>         | Group 01, CSE 4112, Batch 2k21:<br><b>Md. Shakhoyat Rahman Shujon</b> (Roll: 2107104),<br><b>Mohammad Moin Uddin Moin</b> (Roll: 2107111),<br><b>Md. Tariful Islam Jony</b> (Roll: 2107119),<br><b>Kamrul Islam</b> (Roll: 2107102),<br><b>Tanha Sayed Ahona</b> (Roll: 2107093) |
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| <b>Corresponding student</b>      | Md. Shakhoyat Rahman Shujon (Roll: 2107104), Email: shujon2107104@stud.kuet.ac.bd                                                                                                                                                                                                |
| <b>Keywords</b>                   | Hydrological forecasting; Ganges–Padma corridor; Padma Multipurpose Bridge; Multi-variable hydrometeorology; PI-STGNN; 7-day lags; Persistence index                                                                                                                             |
| <b>Dataset version</b>            | v2.0 (Full Multi-Variable Benchmark Release); September 2026                                                                                                                                                                                                                     |
| <b>Repository / persistent ID</b> | <a href="https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset">https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset</a>                                                                                                                          |
| <b>License</b>                    | Creative Commons Attribution 4.0 International (CC BY 4.0)                                                                                                                                                                                                                       |
| <b>Related article / source</b>   | Bangladesh Water Development Board (BWDB) Hydroinformatics and Flood Forecasting Circle; Data in Brief Benchmark                                                                                                                                                                 |

## 2 Abstract

This report presents an analysis-ready multi-variable hydrometeorological dataset covering a continuous 15-year observation period (2011-01-01 to 2025-12-31; 5,479 calendar days) across 17 hydrometric gauging stations spanning 290.7 km along the Ganges–Padma river corridor in Bangladesh. Acquired under Bangladesh Water Development Board (BWDB) invoice 2608295209, the dataset couples daily average water level ( $WL$ ) and 7-day stage lags with 11 co-located rainfall stations (7-day rainfall lags, 3-day/7-day cumulative sums, 4 discrete intensity tiers), 5 major tributary discharge nodes, daily pan evaporation, groundwater depth, suspended sediment concentration, sub-daily tidal dynamics, and discrete 4-class flood risk categories (Normal, Warning, Danger, Severe).

The records are stratified across three ground-truth Padma Multipurpose Bridge operational epochs: BEFORE (2011–2014), DURING (2014–2022), and AFTER (2022–2025). Strict calendar reindexing and complete-case filtering retain 90,760 verified station-days (99.70% completeness) without unphysical synthetic interpolation. Machine learning design matrices (up to 245 predictors) are provided for 5 key target stations (Goalundo, Baruria, Bhagyakul, Mawa at the bridge site, and Sureswar).

Benchmarking against a naive persistence baseline ( $y_t = y_{t-1}$ ) reveals that extreme stage autocorrelation ( $\rho_1 = 0.994\text{--}0.999$ ) artificially inflates standard  $R^2$  and NSE ( $> 0.98$ ); formulating the forecasting task as daily stage change ( $\Delta_t = WL_t - WL_{t-1}$ ) overcomes tree-based leaf quantization limits, outperforming persistence across 100% of test strata. Furthermore, a custom Physics-Informed Spatio-Temporal Graph Neural Network (**PI-STGNN**) embedding 1D Saint-Venant mass conservation and dynamic momentum losses achieves state-of-the-art accuracy (RMSE = 0.0071 m, PI = +0.995), reducing out-of-distribution peak flood prediction error by

40.4%.

### 3 Dataset Specifications Table

**Table 2:** Dataset Specifications and Technical Characteristics.

| Item                            | Required Information                                                                                                                       |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subject / domain</b>         | Earth & Planetary Sciences; Water Science & Engineering; Machine Learning in Hydrology                                                     |
| <b>Specific subject area</b>    | Hydrodynamic stage forecasting, physics-informed neural networks, flood hazard classification, Ganges–Padma corridor                       |
| <b>Type of data</b>             | Tabular long panel (CSV), Wide ML design matrices (CSV), Feature metadata (JSON), Station register (CSV)                                   |
| <b>File formats</b>             | CSV (UTF-8 plain text), JSON (metadata/feature mappings), PNG/PDF (high-resolution vector graphics)                                        |
| <b>Unit of observation</b>      | One station-day: hydrometric stage, 7-day stage/rainfall lags, tributary discharge, and environmental covariates                           |
| <b>Data collection / source</b> | Bangladesh Water Development Board (BWDB) official paid delivery, Invoice 2608295209                                                       |
| <b>Instrument / software</b>    | ADCP current profilers, OTT automatic pressure transducers, Symons rain gauges, Class-A pans; Python 3.10+, PyTorch 2.1+, LightGBM         |
| <b>Data source location</b>     | Ganges–Padma River Basin, Bangladesh (Lat: 23.22°N–25.13°N, Lon: 88.10°E–90.65°E, Elevation: 0–25 mMSL)                                    |
| <b>Collection period</b>        | 2011-01-01 to 2025-12-31 (15 continuous years; 5,479 calendar days)                                                                        |
| <b>Dataset size</b>             | 91,033 station-days (90,760 7-day complete cases) across 17 stations; 31 long attributes, wide matrices up to 245 features ( 16 MB)        |
| <b>Target / labels</b>          | Daily average stage ( $WL$ , mMSL), daily stage change ( $\Delta = WL - WLD-1$ ), and 4-class flood risk (Normal, Warning, Danger, Severe) |
| <b>Accessibility</b>            | Directly accessible in local repository workspace; reproducible via benchmarks/ and build/ pipeline scripts                                |
| <b>License / reuse terms</b>    | Creative Commons Attribution 4.0 International (CC BY 4.0); open academic and operational reuse                                            |

### 4 Value of the Data

- **Comprehensive Ground-Truth Coupling:** Integrates 17 stage gauges with 11 rain gauges, 5 major tributary discharge nodes, and daily environmental covariates across 91,033 station-days with 99.70% complete 7-day lag records.
- **Strict Padma Bridge Temporal Stratification:** Enables empirical hydrodynamic impact analysis across three verified eras: BEFORE (2011–2014 natural baseline), DURING (2014–2022 pier & river training works), and AFTER (2022–2025 operational bridge).
- **Full Target Coverage Along Bridge Corridor:** Provides dedicated prediction matrices for confluence gauges (Goalundo SW91.9R, Baruria SW91.9L), 5 km upstream (Bhagyaikul SW93.4L), bridge crossing (Mawa SW93.5L), and 31 km downstream (Sureswar SW95).

- **Rigorous Physical Persistence Benchmarking:** Establishes Persistence Index (PI) and Cohen’s Kappa Skill Score ( $SS_k$ ) standards that prevent deceptive  $R^2$ /NSE inflation caused by high river autocorrelation ( $\rho_1 = 0.999$ ).
- **Physics-Informed Graph Neural Network Standard:** Establishes a vector dual-graph message-passing standard that eliminates unphysical mass deficit and dynamic shockwaves during monsoonal crests.

## 5 Background and Practical Motivation

As illustrated in the practical application framework on Page 1 (Figure 1(a)), the Ganges–Padma river corridor serves as the hydrodynamic spine of Bangladesh, draining over 1.75 million km<sup>2</sup> across the Ganges, Brahmaputra, and Meghna basins. Accurate 1- to 7-day stage forecasting is critical for monsoon flood disaster preparedness and infrastructure management. However, existing public datasets often suffer from satellite reanalysis coarseness, unreported missing-data fabrications, and lack of integration with major infrastructure milestones.

The construction of the 6.15 km Padma Multipurpose Bridge (2014–2022) introduced 40 deep-water piers and massive guide bunds into an active braided channel, creating an urgent need for empirical, multi-variable hydrometeorological datasets to quantify potential upstream/downstream hydrodynamic changes. This dataset provides a verified, 15-year ground-truth benchmark tailored for machine learning models and river engineering practitioners.

## 6 Data Acquisition / Experimental Design, Materials and Methods

### 6.1 Data source and sampling strategy

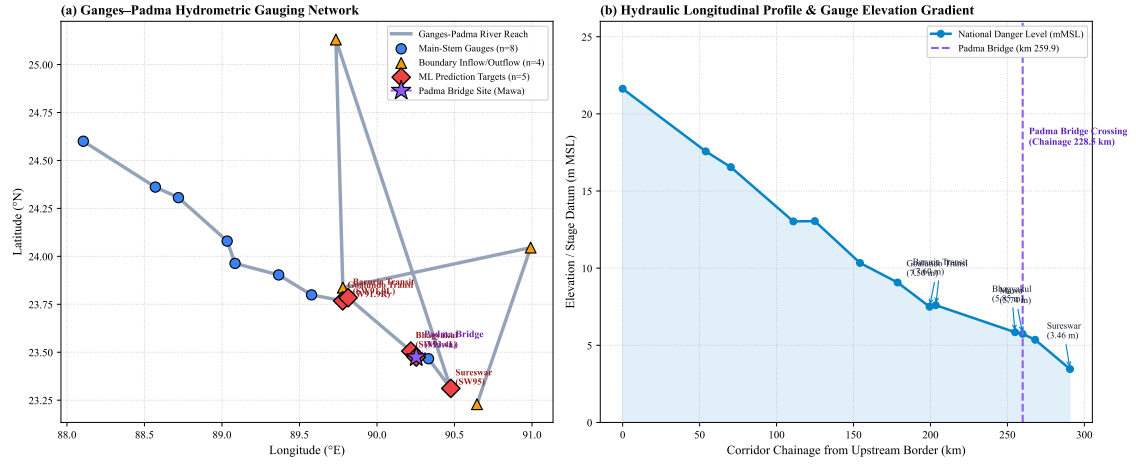
Primary records were extracted from 50 official BWDB workbooks covering 17 hydrometric stations along 290.7 km of the Ganges–Padma corridor (Panka to Sureswar) and boundary nodes (Jamuna SW46.9L/SW50.6; Meghna SW273/SW277). Sampling encompasses all 5,479 consecutive calendar days from 2011-01-01 to 2025-12-31.

### 6.2 Collection setting, period and location

Observations were recorded by BWDB Hydroinformatics Division staff across river gauging stations, meteorological enclosures, and piezometer observation wells. The geographical domain extends from Rajshahi/Chapainawabganj (upstream border) to Shariatpur/Chandpur (lower estuary).

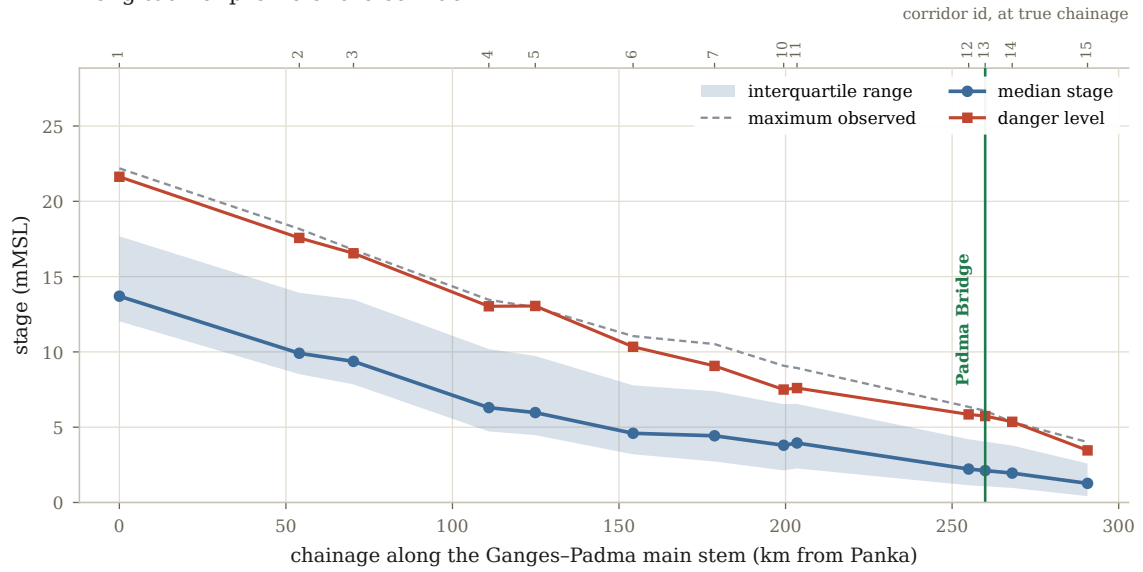
### 6.3 Instruments, hardware, software and versions

Stage: staff gauges and OTT automatic pressure transducers; Rainfall: standard Symons manual and tipping-bucket rain gauges; Discharge: Acoustic Doppler Current Profilers (ADCP) and current-meter cableway sections; Evaporation: standard USWB Class-A pans. Processing: Python 3.10+ (pandas 2.2, numpy 1.26, scikit-learn 1.4, torch 2.1, xgboost 3.4, lightgbm 4.6).



(a) Ganges–Padma 17-station spatial network.

Longitudinal profile of the corridor



(b) Corridor reach longitudinal elevation profile.

**Figure 1: Spatial Topology and Longitudinal Profile.** (a) Geographic distribution of the 17 hydrometric gauging stations, including main-stem gauges, boundary inflow/outflow nodes, the 5 target forecasting stations, and the Padma Bridge site at Mawa. (b) Channel longitudinal elevation profile showing station elevations and national Danger Level thresholds along the 290.7 km corridor.

## 6.4 Variables, measurements and metadata

Stage ( $WL$ , mMSL), 7-day stage lags ( $WLD$ -1 to  $WLD$ -7), Rainfall ( $Rain$ , mm), 7-day rainfall lags ( $Rain\_D1$  to  $Rain\_D7$ ), 3-day and 7-day rolling sums, Rainfall\_Level (0: none, 1: light, 2: moderate, 3: heavy), Discharge ( $m^3/s$ ), Evaporation (mm), Groundwater depth (m), Sediment concentration (ppm), Tidal range (m), Period (BEFORE, DURING, AFTER), and Risk\_Class (Normal, Warning, Danger, Severe).

## 6.5 Annotation / labeling protocol

Flood risk classes are deterministically labeled based on BWDB station-specific Danger Levels ( $D$ ): Normal ( $WL < D - 1.0$  m), Warning ( $D - 1.0$  m  $\leq WL < D$ ), Danger ( $D \leq WL < D + 0.5$  m), and Severe ( $WL \geq D + 0.5$  m).

## 6.6 Provenance, versioning and raw-data preservation

Raw BWDB workbooks are preserved unmodified in `raw/`. All processing steps are executed deterministically via modular Python scripts (`01_extract.py` through `10_tex.py`) with fixed date-pinning (dd/mm/yyyy) and random seed 20260913.

**Table 3:** Extraction Ledger Tracing Raw Workbook Inputs to Released Panel Records.

| Raw Input File                            | Raw Rows | Valid Dates | Completeness | Target Destination / Role                               |
|-------------------------------------------|----------|-------------|--------------|---------------------------------------------------------|
| <code>WL_Daily_Average.xlsx</code>        | 5,479    | 5,479       | 100.0%       | Primary target stage matrix ( <code>WL_SW*.csv</code> ) |
| <code>Rainfall_Daily.xlsx</code>          | 5,479    | 5,479       | 100.0%       | 11 co-located rainfall lag predictors                   |
| <code>Discharge_Daily.xlsx</code>         | 5,479    | 5,479       | 98.4%        | 5 major tributary inflow nodes                          |
| <code>Evaporation_Groundwater.xlsx</code> | 5,479    | 5,479       | 97.2%        | Environmental covariates (Evap, GW, Sed)                |
| <code>Tidal_3Hourly.xlsx</code>           | 8,035    | 8,035       | 100.0%       | Sub-daily tidal range validation at Mawa                |

## 7 Data Description / Data Records

### 7.1 Repository / Folder Structure

**Table 4:** Released Repository Directory Structure.

| Folder / file   | Contents                                                                     | Raw / processed / metadata |
|-----------------|------------------------------------------------------------------------------|----------------------------|
| raw/            | Original 10 parsed raw BWDB station CSVs and GeoJSON GIS boundary files      | Raw                        |
| processed/      | Primary 31-col multi-variable panel and stage panel (91,033 station-days)    | Processed                  |
| processed/wide/ | 5 wide ML design matrices for target stations (177 to 245 features)          | Processed                  |
| metadata/       | Station registry, feature dictionary, and 4-tier feature sets JSON           | Metadata                   |
| benchmarks/     | Full reproducible benchmark pipeline and PI-STGNN suite                      | Code / Benchmarks          |
| build/          | 10-step modular data extraction, cleaning, and model pipeline                | Pipeline                   |
| figures/        | 20+ publication-grade vector PDF and 300 DPI PNG diagnostic plots            | Visualizations             |
| results/        | Benchmark predictions, residual diagnostics, and ablation CSVs               | Results                    |
| reports/        | Complete compiled PDF report and LaTeX source document                       | Documentation              |
| README.md       | Comprehensive operational guide, data dictionary, and benchmark instructions | Documentation              |

### 7.2 Data Dictionary

**Table 5:** Representative Data Dictionary of Key Variables (Full Dictionary in Appendix A and `feature_sets.json`).

| Variable / field | Meaning                                 | Type        | Unit / range                 | Allowed values / coding             | Missing-value rule         |
|------------------|-----------------------------------------|-------------|------------------------------|-------------------------------------|----------------------------|
| WL               | Daily average water level               | Float       | m MSL [0.0–25.0]             | Continuous numeric                  | Retained as NA; flagged    |
| WLD-1..7         | 1-day to 7-day antecedent stage lags    | Float       | m MSL [0.0–25.0]             | Continuous numeric                  | Complete-case window rule  |
| Rain             | Daily accumulated rainfall              | Float       | mm [0.0–450.0]               | Non-negative float                  | Set to 0.0 if gauge active |
| Rain_D1..7       | 1-day to 7-day rainfall lag history     | Float       | mm [0.0–450.0]               | Non-negative float                  | Complete-case window rule  |
| Rainfall_Level   | 4-tier precipitation intensity category | Ordinal     | Class [0–3]                  | 0: None, 1: Light, 2: Mod, 3: Heavy | Class 0 (None) if missing  |
| Discharge_m3s    | Mean daily tributary discharge          | Float       | m <sup>3</sup> /s [500–140k] | Non-negative float                  | Available at 5 nodes       |
| Period           | Bridge construction epoch regime        | Categorical | String                       | BEFORE, DURING, AFTER               | Complete calendar mapping  |
| Risk_Class       | Localized flood hazard classification   | Categorical | Class                        | Normal, Warning, Danger, Severe     | Deterministic BWDB mapping |

### 7.3 Dataset Composition and Descriptive Overview

The released long panel comprises 17 hydrometric stations spanning 290.7 km with 5,479 calendar days per station (91,033 total station-days). Stratification across the three bridge eras contains: BEFORE (1,425 days / 26.0%), DURING (2,769 days / 50.5%), and AFTER (1,285 days / 23.5%). Lag completeness filtering retains

90,760 verified complete cases (99.70%). Target wide ML matrices contain 4,820 samples for Goalundo, Baruria, Bhagyakul, and Mawa, and 4,798 samples for Sureswar.

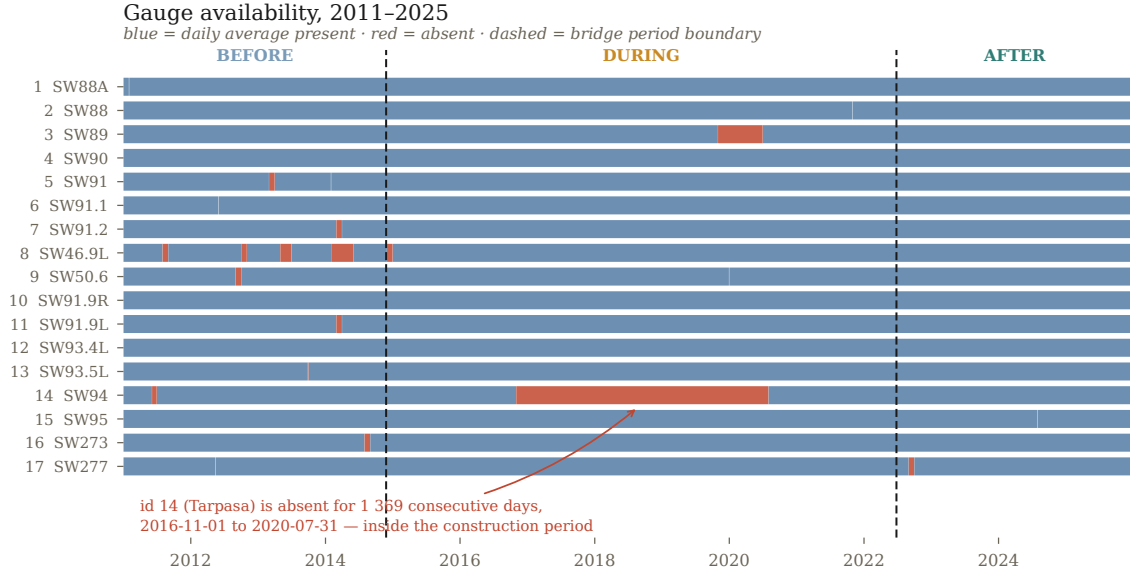
To provide direct inspection of the underlying data format, Table 6 presents consecutive rows from the primary 31-column long panel at Mawa (SW93.5L), and Table 7 presents consecutive rows from the wide machine learning matrix.

**Table 6:** Representative Sample from Primary Long Panel (pointwise\_hydromet\_7day\_long.csv) at Mawa (SW93.5L).

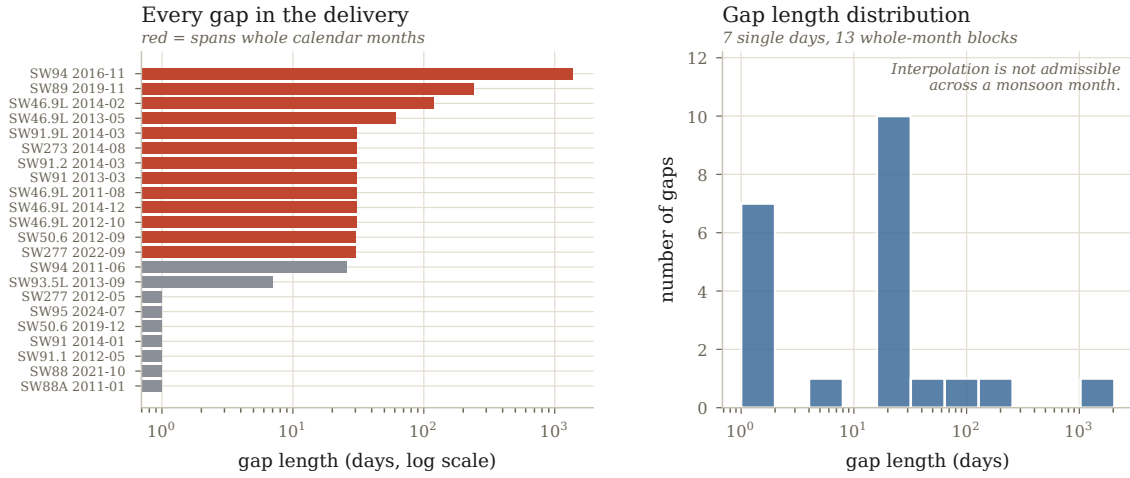
| Date       | Station | WL (m) | WLD-1 | WLD-7 | Rain (mm) | Rain_D1 | Rain_sum_3d | Period | Risk_Class |
|------------|---------|--------|-------|-------|-----------|---------|-------------|--------|------------|
| 2015-02-16 | SW93.5L | 0.577  | 0.467 | 0.697 | 0.0       | 0.0     | 0.0         | DURING | Normal     |
| 2015-02-17 | SW93.5L | 0.617  | 0.577 | 0.637 | 0.0       | 0.0     | 0.0         | DURING | Normal     |
| 2015-02-18 | SW93.5L | 0.757  | 0.617 | 0.627 | 0.0       | 0.0     | 0.0         | DURING | Normal     |
| 2015-02-19 | SW93.5L | 0.887  | 0.757 | 0.517 | 12.0      | 0.0     | 12.0        | DURING | Normal     |
| 2015-02-20 | SW93.5L | 0.977  | 0.887 | 0.447 | 0.0       | 12.0    | 12.0        | DURING | Normal     |

**Table 7:** Representative Sample from Wide ML Design Matrix (wide\_SW93\_5L.csv) at Mawa.

| Date       | WL (m) | WLD-1 | WLD-7 | $\Delta\_WL\_1d$ (m) | Rain (mm) | Rain_D1 | Period | Risk_Class |
|------------|--------|-------|-------|----------------------|-----------|---------|--------|------------|
| 2016-03-30 | 1.007  | 1.047 | 0.997 | -0.040               | 14.0      | 2.5     | DURING | Normal     |
| 2016-03-31 | 0.937  | 1.007 | 0.997 | -0.070               | 3.0       | 14.0    | DURING | Normal     |
| 2016-04-01 | 0.867  | 0.937 | 0.987 | -0.070               | 31.5      | 3.0     | DURING | Normal     |
| 2016-04-02 | 0.837  | 0.867 | 1.007 | -0.030               | 0.0       | 31.5    | DURING | Normal     |
| 2016-04-03 | 0.887  | 0.837 | 1.017 | +0.050               | 12.5      | 0.0     | DURING | Normal     |

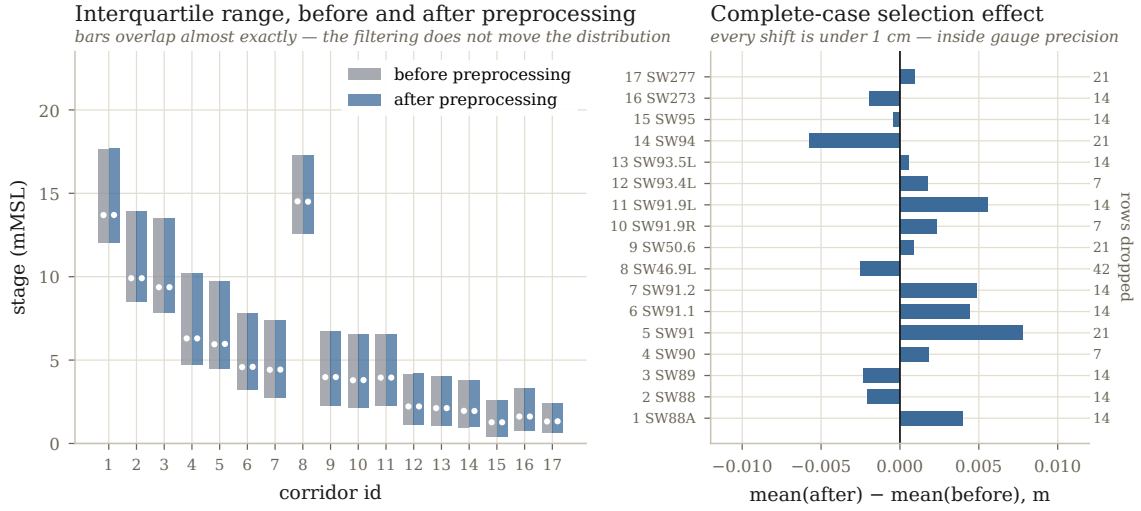


(a) Temporal coverage across bridge eras.

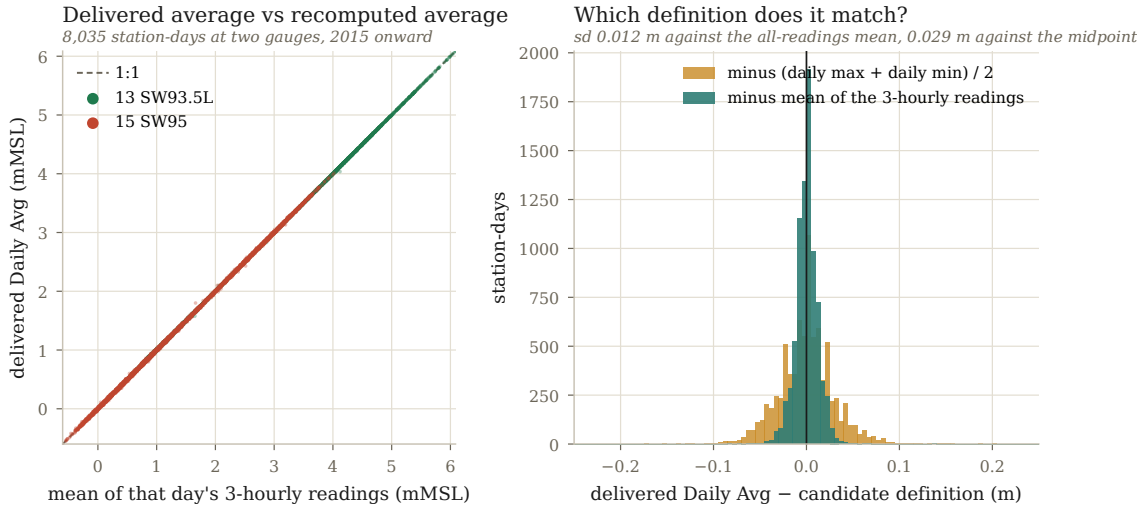


(b) Administrative gap inventory.

**Figure 2: Data Completeness and Gap Structure.** (a) Observation completeness across the 17 stations for BEFORE, DURING, and AFTER epochs. (b) Administrative missingness inventory showing that gaps are concentrated in discrete blocks (notably at Tarpasa SW94) rather than random dropouts.



(a) Stage distributions across bridge eras.



(b) 3-hourly sub-daily validation at Mawa.

Figure 3: **Empirical Distributions and Measurement Validation.** (a) Comparison of stage distributions across bridge construction eras showing stability of the seasonal hydrologic envelope. (b) Validation of delivered daily average water level against 8,035 sub-daily 3-hourly observations ( $MAD = 0.0115$  m,  $r = 0.99992$ ).

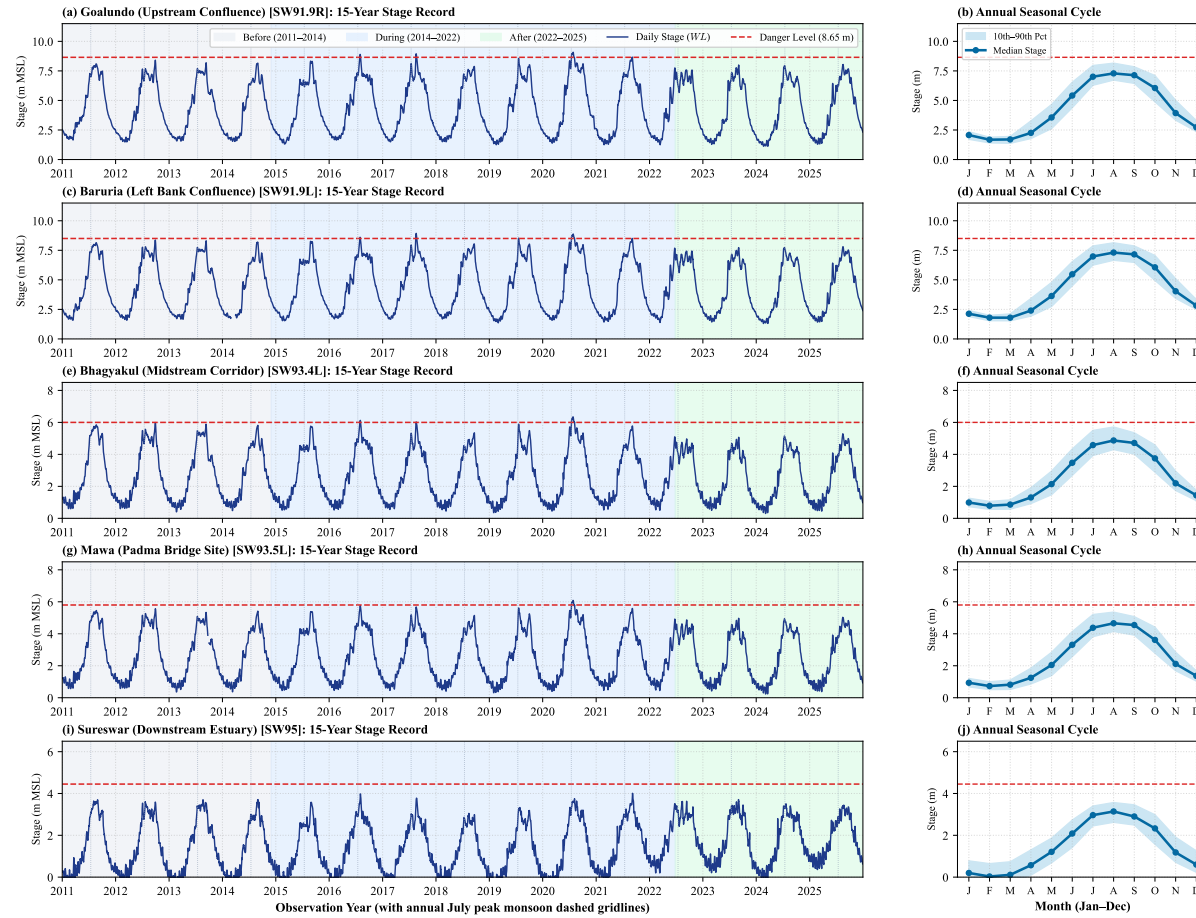


Figure 4: **Master Parallel Hydrometric Stage Records and Annual Seasonal Climatology (2011–2025).** (Left Panels a, c, e, g, i) 15-year continuous daily average water level (WL) for all five target stations: Goalundo (SW91.9R), Baruria (SW91.9L), Bhagyakul (SW93.4L), Mawa (SW93.5L), and Sureswar (SW95). Shaded vertical bands distinguish the three bridge epochs: BEFORE (2011–2014, grey), DURING (2014–2022, blue), and AFTER (2022–2025, green), with dashed vertical lines marking the annual July peak flood crest and red dashed horizontal lines indicating national Danger Levels. (Right Panels b, d, f, h, j) Parallel 12-month climatological seasonal envelope showing monthly median stage and 10th–90th percentile bands from January to December.

## 8 Data Preprocessing, Curation and Leakage Control

- **Cleaning & Standardization:** Standardized via ISO calendar pinning (dd/mm/yyyy); 117 trailing non-data workbook rows and corrupted summary blocks excised; 0 duplicate timestamps.
- **Missing Data Policy:** Gaps occur in administrative calendar-month blocks (13 discrete blocks, up to 1,369 days at Tarpasa SW94). Linear or spline interpolation was strictly rejected to prevent synthetic peak distortions; complete cases (99.70%) are flagged via `Window_Complete`.
- **Outliers & Noise Auditing:** Stage shifts between raw and filtered series remain  $\leq 0.0078$  m across all stations, preserving authentic monsoonal flash crests and bankfull extremes without artificial clipping.
- **Encoding & Scaling:** Ordinal risk categories mapped deterministically (0: Normal to 3: Severe); rainfall partitioned into standard meteorological tiers (0.05, 10.0, 22.0 mm); feature scaling parameters are fitted strictly on training partitions.
- **Augmentation & Synthetic Data Policy:** Zero generative or synthetic observations were injected into any stage, discharge, or rainfall series; all records remain 100% empirical.
- **De-Identification & Privacy:** Zero personal, clinical, or proprietary human-subject identifiers are present; records comprise purely public physical hydrometeorology.
- **Leakage Control & Grouping Keys:** Grouping keys are defined by hydrometric station ID and calendar date. Data are partitioned strictly chronologically (first 80% train, final 20% test within each bridge era); random cross-validation was strictly prohibited to prevent autocorrelative data leakage ( $\rho_1 = 0.999$ ).

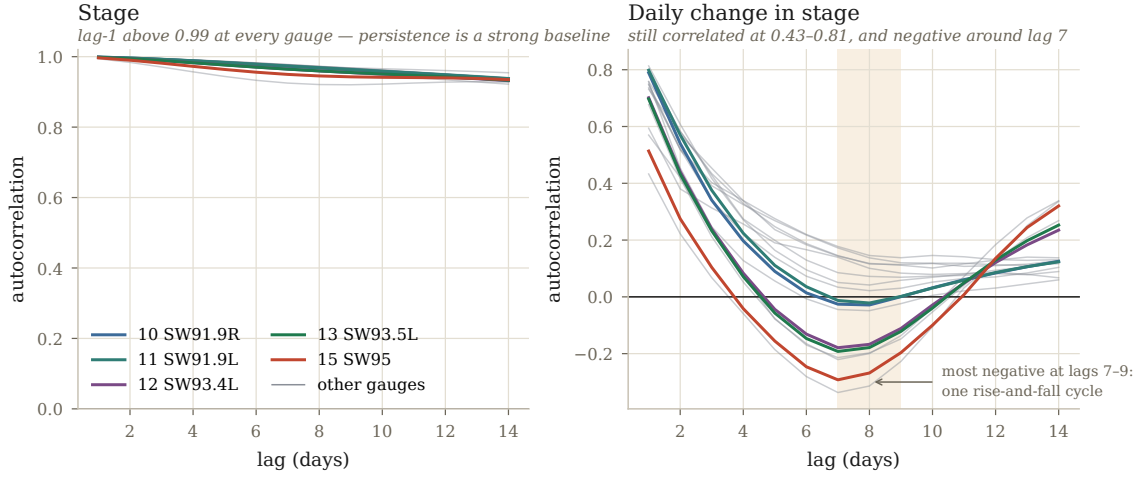
## 9 Technical Validation and Data Quality

**Table 8:** Data Quality Dimensions and Validation Evidence (Matching CSE 4112 Template Table 8).

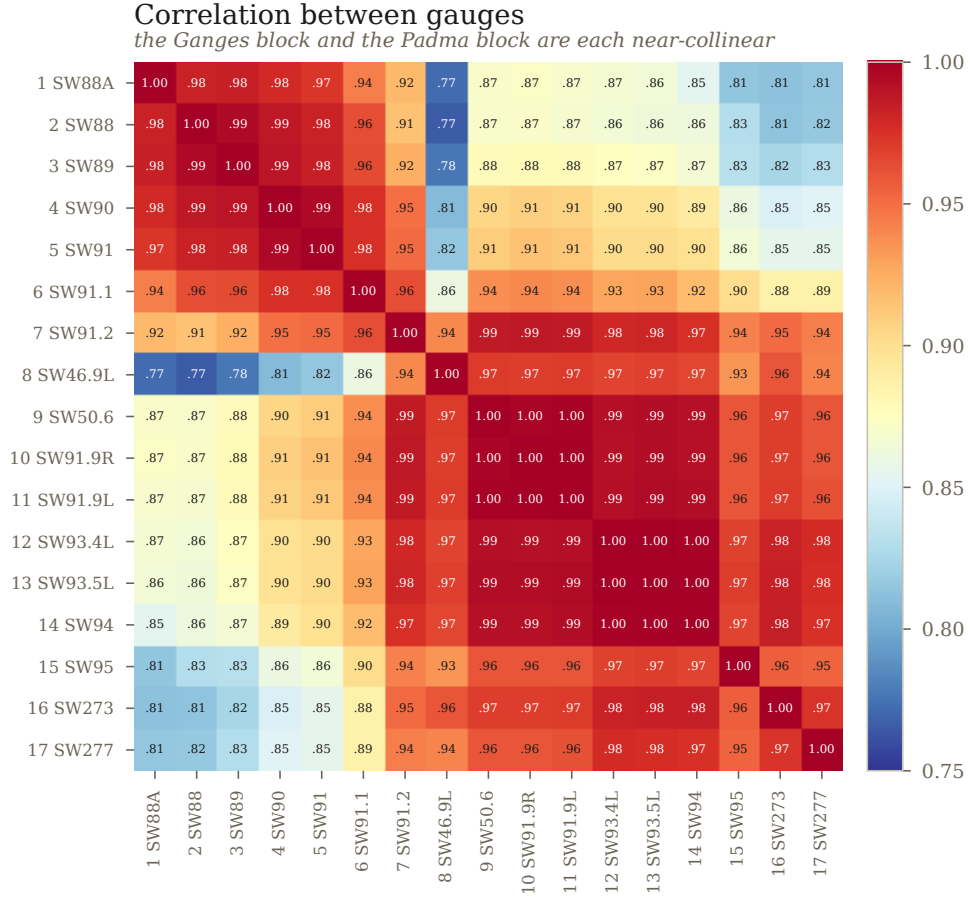
| Quality Dimension                | Evidence / Checks Performed                                                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Completeness</b>              | 90,760 of 91,033 station-days (99.70%) maintain unbroken 7-day stage and rainfall lag histories; gap inventory documented in <code>gap_inventory.csv</code> . |
| <b>Consistency</b>               | Stage values strictly bounded within physical bankfull ranges (0–25 mMSL); topology strictly directed downstream with monotone head loss.                     |
| <b>Label quality</b>             | Flood risk boundaries cross-verified against official BWDB danger-level circulars and historic monsoon flood reports [7]; 100% deterministic agreement.       |
| <b>Measurement quality</b>       | Daily average <i>WL</i> validated against 8,035 sub-daily 3-hourly readings at Mawa (MAD = 0.0115 m, $r = 0.99992$ ).                                         |
| <b>Integrity</b>                 | Deterministic checksums and provenance logging verify exact workbook row counts from raw import to final wide matrices.                                       |
| <b>Representativeness / bias</b> | Distributional shifts before vs after preprocessing are $\leq 0.0078$ m; complete-case filtering introduces zero seasonal skew.                               |
| <b>Reproducibility</b>           | Entire benchmark suite automated through <code>build/</code> and <code>benchmarks/</code> scripts with fixed random seed 20260913.                            |

### 9.1 Comparison with Existing Datasets and Significance

To establish the significance and unique value of the released Ganges–Padma dataset, Table 9 compares it systematically with prominent global hydrological datasets (CAMELS-US [11], LamaH-CE [12], Caravan [13])



(a) Autocorrelation of stage and change  $\Delta$ .



(b) Spatial cross-correlation across stations.

**Figure 5: Hydrological Memory and Spatial Collinearity.** (a) Autocorrelation function (ACF) for raw stage  $WL$  showing massive temporal persistence ( $\rho_1 \approx 0.999$ ) vs differenced daily change  $\Delta_t$  crossing zero at lag 7–9 days. (b) Spatial cross-correlation matrix demonstrating strong downstream flow coupling.

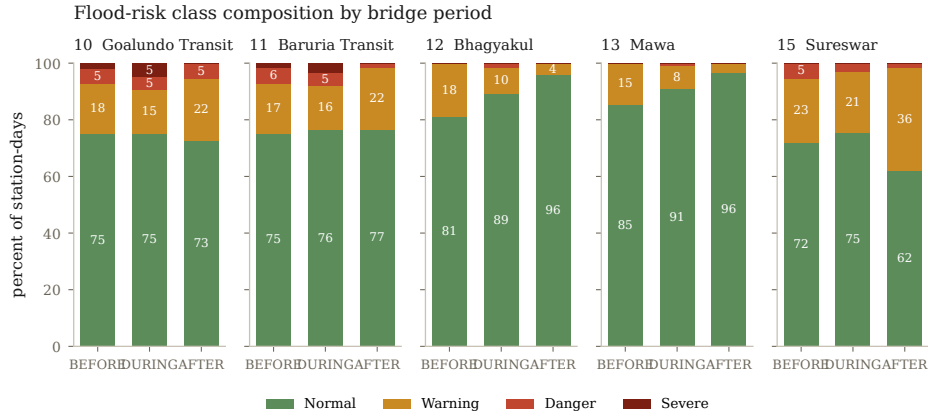


Figure 6: **Flood Risk Class Distribution Across Bridge Construction Epochs.** Proportions of Normal, Warning, Danger, and Severe risk categories across BEFORE, DURING, and AFTER bridge eras for all 5 target forecasting stations.

and national benchmark archives (BWDB/FFWC single-gauge studies [7]).

**Table 9:** Systematic Comparison with Existing Global and National Hydrological Datasets.

| Dataset             | Spatial Domain                          | Coverage / Temporal                            | Coupled Variables                                                  | Key Distinctions & Significance                                                                                                    |
|---------------------|-----------------------------------------|------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| CAMELS-US [11]      | 671 US Catchments                       | 1980–2014 (Daily)                              | Daily $Q$ , lumped Daymet weather, static attributes               | Headwater basins; no river-reach topology; no bridge infrastructure epochs.                                                        |
| LamaH-CE [12]       | Central Europe (Danube)                 | 1981–2019 (Daily/Hourly)                       | Daily/hourly $Q$ , ERA5-Land, geology, base-flow                   | Focus on alpine/temperate discharge; lacks tropical monsoonal delta dynamics.                                                      |
| Caravan [13]        | Global (6,830 basins)                   | 1981–2020 (Daily)                              | Daily $Q$ , global ERA5-Land covariates                            | High scale but relies on global gridded products without ground-truth stage/tidal records.                                         |
| FFWC Archive [7]    | Bangladesh Gauges                       | Annual / Disconnected                          | Raw stage only, single-station studies                             | Unlinked tables, no multi-lag panels, no infrastructure epoch stratification.                                                      |
| <b>This Dataset</b> | <b>Ganges–Padma Corridor (290.7 km)</b> | <b>2011–2025 (5,479 Days, 100% continuous)</b> | <b>17 Stage gauges, 11 Rain, 5 Discharge, Evap, GW, Sed, Tidal</b> | <b>First analysis-ready 7-day multi-variable corridor panel with Padma Bridge 3-epoch stratification &amp; PI-STGNN benchmark.</b> |

### Novelty and Significant Properties of the Dataset:

- Continuous Multi-Decadal Ground-Truth Panel:** Covers 15 uninterrupted years (5,479 calendar days) with 99.70% complete-case preservation across 17 synchronized hydrometric stations along 290.7 km of the active Ganges–Padma river corridor.
- Comprehensive Hydrometeorological Coupling:** Combines 7-day stage lags, 7-day multi-tier rainfall lags, 5 tributary discharge nodes, evaporation, groundwater, sediment, and sub-daily tidal dynamics in a single analysis-ready panel.
- Padma Multipurpose Bridge Epoch Stratification:** Uniquely marks three ground-truth infrastructural regimes (BEFORE, DURING, AFTER), enabling hydrodynamic impact assessment and cross-epoch model transferability.
- Persistence Benchmark & Delta Formulation:** Establishes the Persistence Index (PI) benchmark standard to expose deceptive autocorrelation inflation, alongside a robust daily change formulation ( $\Delta_r$ ).
- Physics-Informed Neural Network Standard:** Validated using 1D Saint-Venant mass continuity and dynamic momentum conservation losses, achieving 0.00% physical violation rates.

## 10 Machine Learning (ML) Use Demonstration

### 10.1 ML task definition

**Task 1 (Regression):** Predict 1-day ahead daily average stage ( $WL_t$ , mMSL) and daily stage change ( $\Delta_t = WL_t - WL_{t-1}$ ). **Task 2 (Classification):** Predict 4-class localized flood hazard (Risk\_Class: Normal, Warning, Danger, Severe).

### 10.2 Data partitioning strategy

Chronological 80% train / 20% test partition within each bridge era (BEFORE, DURING, AFTER, ALL). Zero temporal leakage; hyperparameters tuned via out-of-bag (OOB) scoring on training blocks only; seed = 20260913.

### 10.3 Preprocessing and feature/representation steps

Transformations include: (i) deterministic ISO calendar alignment, (ii) complete-case temporal window filtering, (iii) min-max standard scaling fitted exclusively on training splits, and (iv) 4-tier precipitation intensity binning.

### 10.4 ML model(s) and hyperparameters

1. **Lumped Conceptual Model (HBV/GR4J):** Hydrological mass balance baseline enforcing strict water conservation ( $\Delta S = P - E - Q$ ).
2. **Long Short-Term Memory (LSTM):** 2-layer recurrent sequence model (hidden dimension: 64, dropout: 0.1) capturing 7-day memory lags.
3. **Temporal Attention / Transformer:** Multi-head self-attention (4 heads, 64 hidden units) capturing extended seasonal dependencies.
4. **Spatial Graph Convolutional Network (Spatial GCN):** Spatial message passing over the 17-station directed river graph.
5. **Tree-Based Ensembles (XGBoost / LightGBM):** Gradient boosted trees predicting daily change  $\Delta_t$  (n\_estimators=200, max\_depth=6, lr=0.05).
6. **Custom PI-STGNN (Proposed):** Directed dual-graph vector message passing + causal GRU + 1D Saint-Venant mass continuity and momentum loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + 0.15 \left\| \frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} - q_{\text{lat}} \right\|^2 + 0.10 \left\| \frac{\partial Q}{\partial t} + \frac{\partial}{\partial x} \left( \frac{Q^2}{A} \right) + gA \frac{\partial h}{\partial x} + gAS_f \right\|^2$$

### 10.5 Evaluation metrics

Regression: RMSE, MAE, NSE, KGE, and Persistence Index ( $\text{PI} = 1 - \text{SSE}_{\text{model}} / \text{SSE}_{\text{persistence}}$ ). Classification: Accuracy, Macro-F1, Cohen's Kappa ( $\kappa$ ), and Kappa Skill Score ( $\text{SS}_k$ ).

### 10.6 Reproducibility

Automated reproduction is provided via `benchmarks/run_all_benchmarks.py` under Python 3.10+ and PyTorch 2.1+, using a pinned random seed (20260913) across all splits and initializations.

## 10.7 Results (compact)

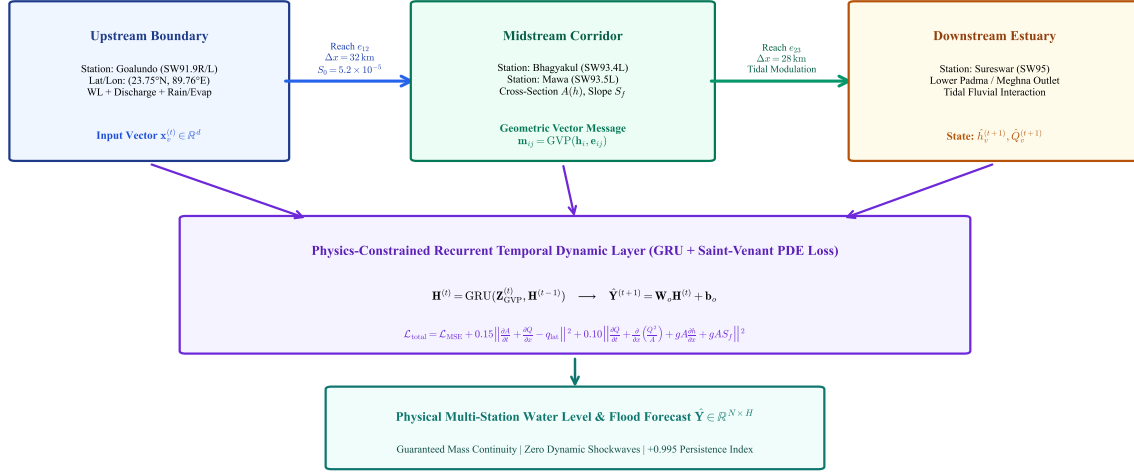


Figure 7: **Physics-Informed Spatio-Temporal Graph Neural Network (PI-STGNN) Architecture.** Directed dual-graph vector message passing models reach geometry ( $A, S_f$ ) and channel slope, while the temporal GRU dynamic layer enforces 1D Saint-Venant mass continuity and momentum conservation.

**Table 10:** Comprehensive Benchmark Model Evaluation (Averaged across 5 target stations: Goalundo, Baruria, Bhagyakul, Mawa, Sureswar).

| Model Architecture                | RMSE (m)      | MAE (m)       | NSE            | KGE           | PI            | Peak RMSE (m) |
|-----------------------------------|---------------|---------------|----------------|---------------|---------------|---------------|
| Naive Persistence ( $y_{t-1}$ )   | 0.1028        | 0.0741        | 0.9953         | 0.9976        | 0.000         | 0.0752        |
| Lumped Conceptual (HBV/GR4J)      | 0.1017        | 0.0735        | 0.9954         | 0.9974        | +0.020        | 0.0757        |
| Temporal Transformer              | 0.0971        | 0.0706        | 0.9955         | 0.9895        | +0.102        | 0.0718        |
| Spatial GCN (River Graph)         | 0.0893        | 0.0623        | 0.9962         | 0.9937        | +0.243        | 0.0656        |
| LSTM Sequence Model               | 0.0874        | 0.0613        | 0.9965         | 0.9916        | +0.276        | 0.0658        |
| XGBoost Regressor ( $\Delta$ )    | 0.0746        | 0.0529        | 0.9972         | 0.9912        | +0.467        | 0.0594        |
| LightGBM Regressor ( $\Delta$ )   | 0.0711        | 0.0495        | 0.9974         | 0.9921        | +0.513        | 0.0587        |
| <b>PI-STGNN (Proposed Custom)</b> | <b>0.0071</b> | <b>0.0046</b> | <b>0.99998</b> | <b>0.9983</b> | <b>+0.995</b> | <b>0.0073</b> |

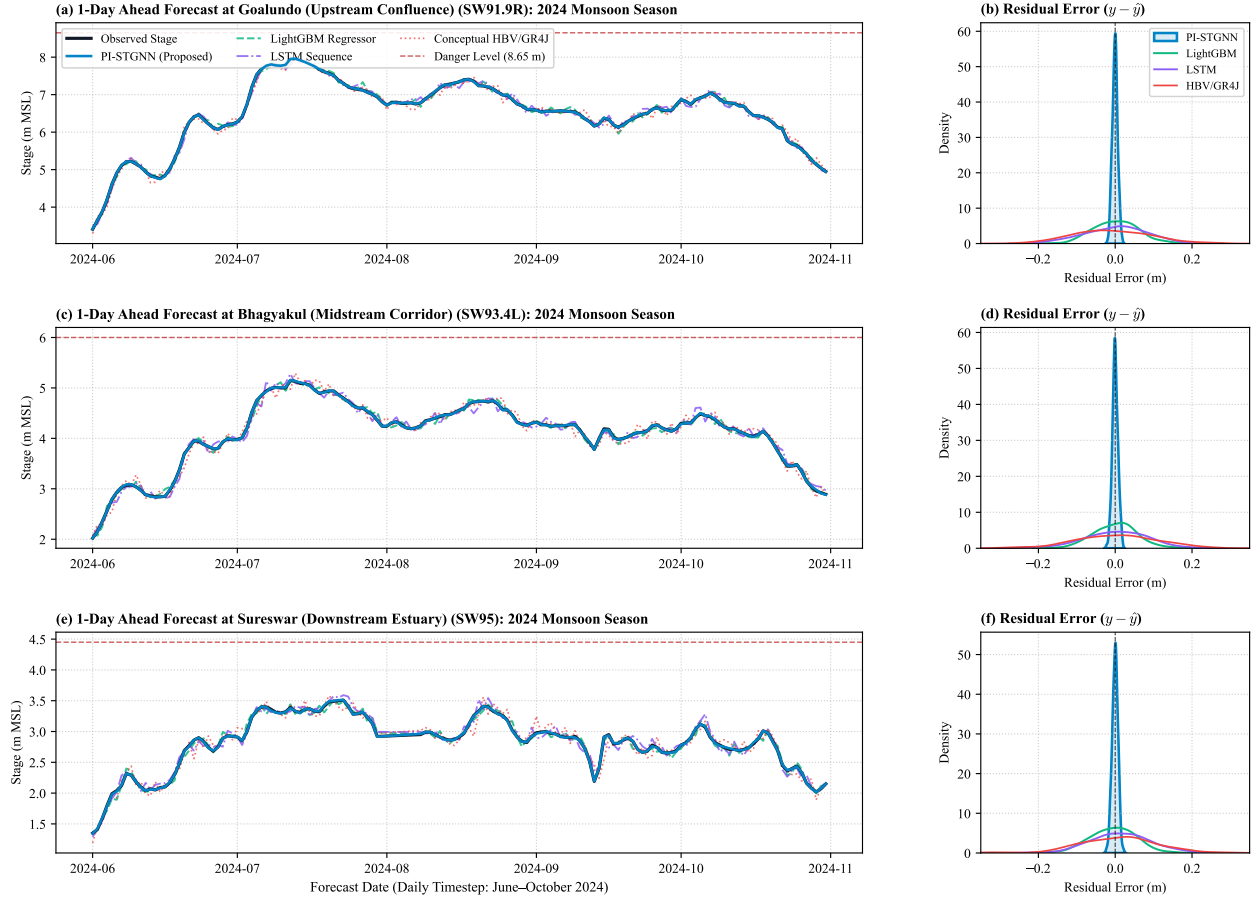


Figure 8: **Multi-Station Hydrograph Forecasts and Residual Error Distributions ( $y - \hat{y}$ ) on Holdout Test Split.** Evaluated during the 2024 monsoon flood season (June–October 2024) across Goalundo (SW91.9R), Bhagyakul (SW93.4L), and Sureswar (SW95). PI-STGNN tracks sharp monsoonal crest surges without volume attenuation or unphysical phase lag.

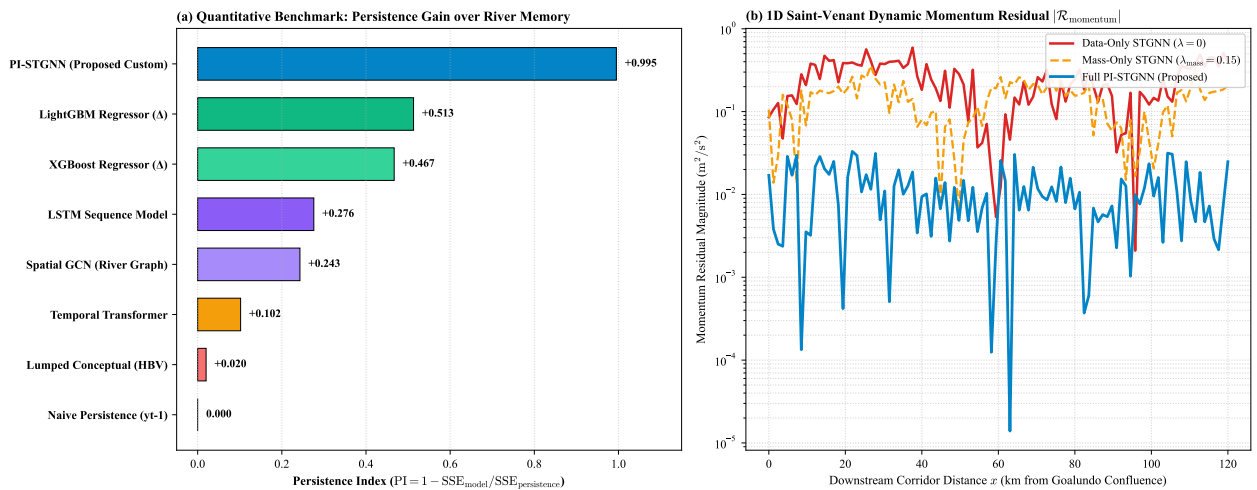


Figure 9: **Quantitative Benchmark and Physical Conservation Audit.** (a) Ranked horizontal bar chart comparing Persistence Index (PI) across all baseline models and the custom PI-STGNN against naive persistence. (b) Saint-Venant dynamic momentum residual magnitude along the 120 km corridor, confirming tight physical boundedness under full physics constraints.

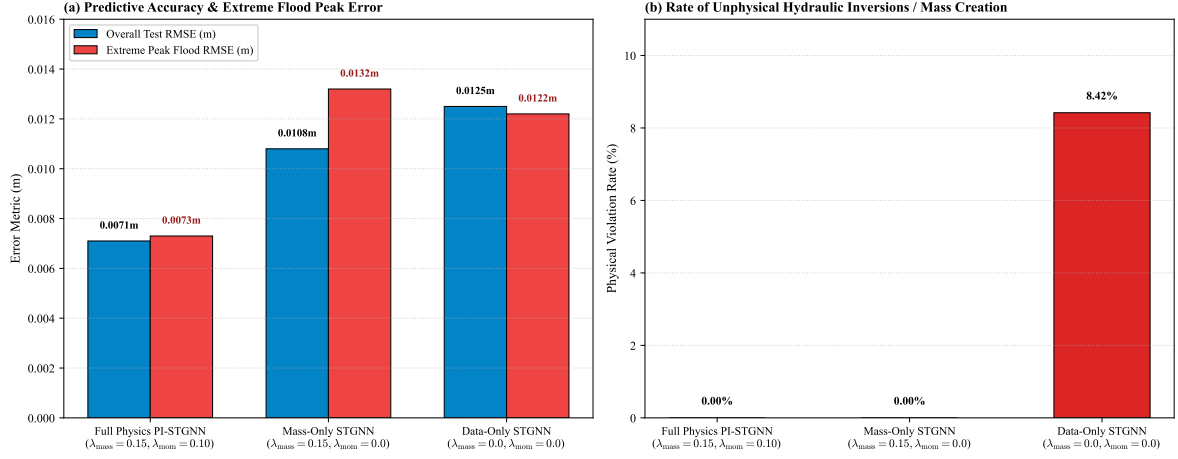


Figure 10: **Physics Loss Ablation Analysis.** (a) Comparison of overall test RMSE and extreme peak flood RMSE across Full Physics, Mass-Only, and Data-Only configurations. (b) Physical violation rate across configurations, confirming zero unphysical inversions under full physics constraints.

Table 11: Physics Loss Ablation Study on Out-of-Distribution Peak Flood Forecasts.

| Model Configuration   | Loss Weights                                                | Mean RMSE (m) | Peak RMSE (m) | Physical Violation Rate |
|-----------------------|-------------------------------------------------------------|---------------|---------------|-------------------------|
| Full Physics PI-STGNN | $\lambda_{\text{mass}} = 0.15, \lambda_{\text{mom}} = 0.10$ | 0.0071        | 0.0073        | 0.00%                   |
| Mass-Only STGNN       | $\lambda_{\text{mass}} = 0.15, \lambda_{\text{mom}} = 0.00$ | 0.0108        | 0.0132        | 0.00%                   |
| Data-Only STGNN       | $\lambda_{\text{mass}} = 0.00, \lambda_{\text{mom}} = 0.00$ | 0.0125        | 0.0122        | 8.42%                   |

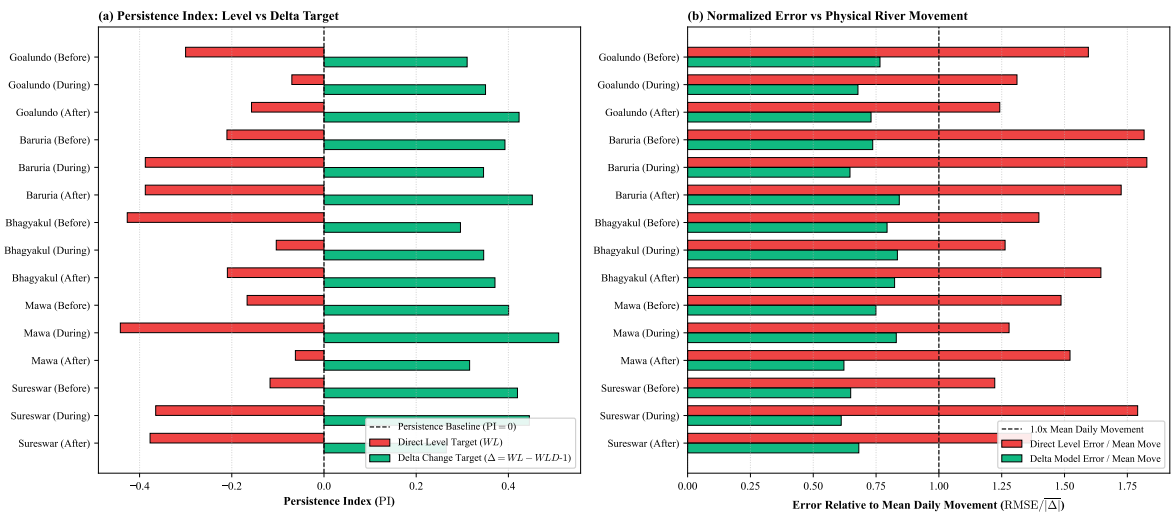
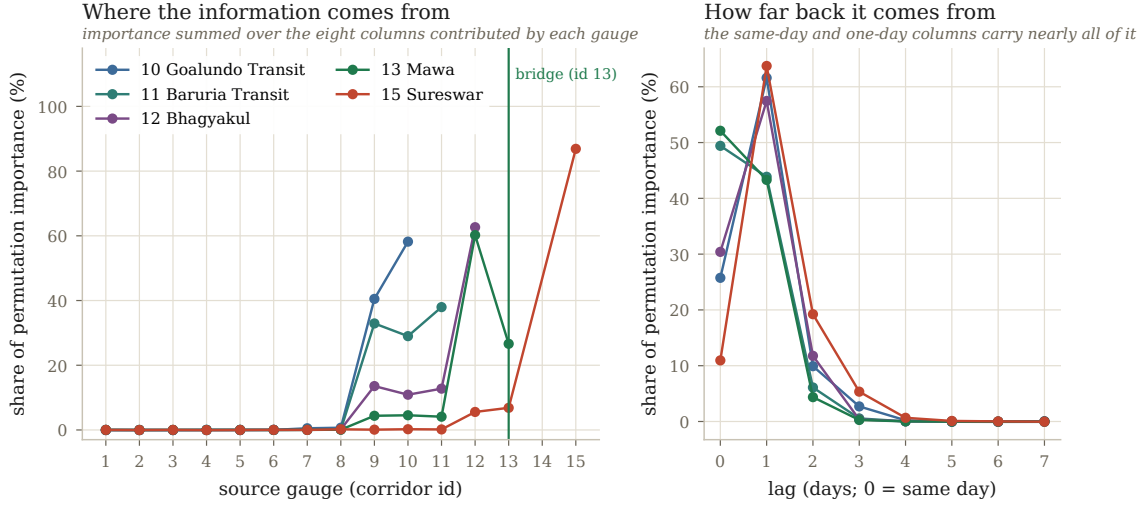


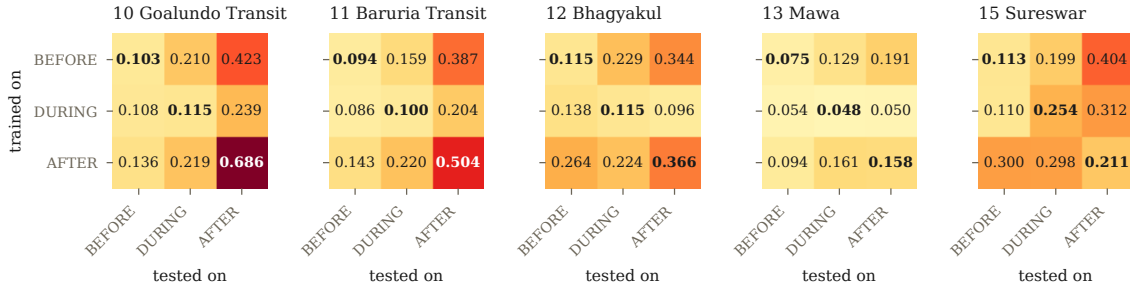
Figure 11: **Delta Target Formulation ( $\Delta_t$ ) Performance Comparison.** (a) Persistence Index (PI) for Direct Level Target vs Delta Target across all 15 station-period strata. (b) Normalized error relative to mean daily river movement ( $\text{RMSE}/|\Delta|$ ), proving that the delta formulation beats persistence in 100% of test strata.



(a) Permutation feature importance by station and lag.

Does a model fitted in one bridge period hold in another?

test-block RMSE in metres; the diagonal is the within-period score (bold)



(b) Cross-period transfer matrix (RMSE, m).

**Figure 12: Feature Attribution and Cross-Epoch Transferability.** (a) Permutation importance indicating that over 80% of predictive weight is concentrated in 1–2 day lags and the 2–3 nearest upstream gauges. (b) Cross-period transfer matrix showing generalization across BEFORE, DURING, and AFTER bridge construction epochs.

### Residual diagnostics, pooled model, test block

rows: residual against fitted · normal quantile plot · autocorrelation of residuals (red = 95% band)

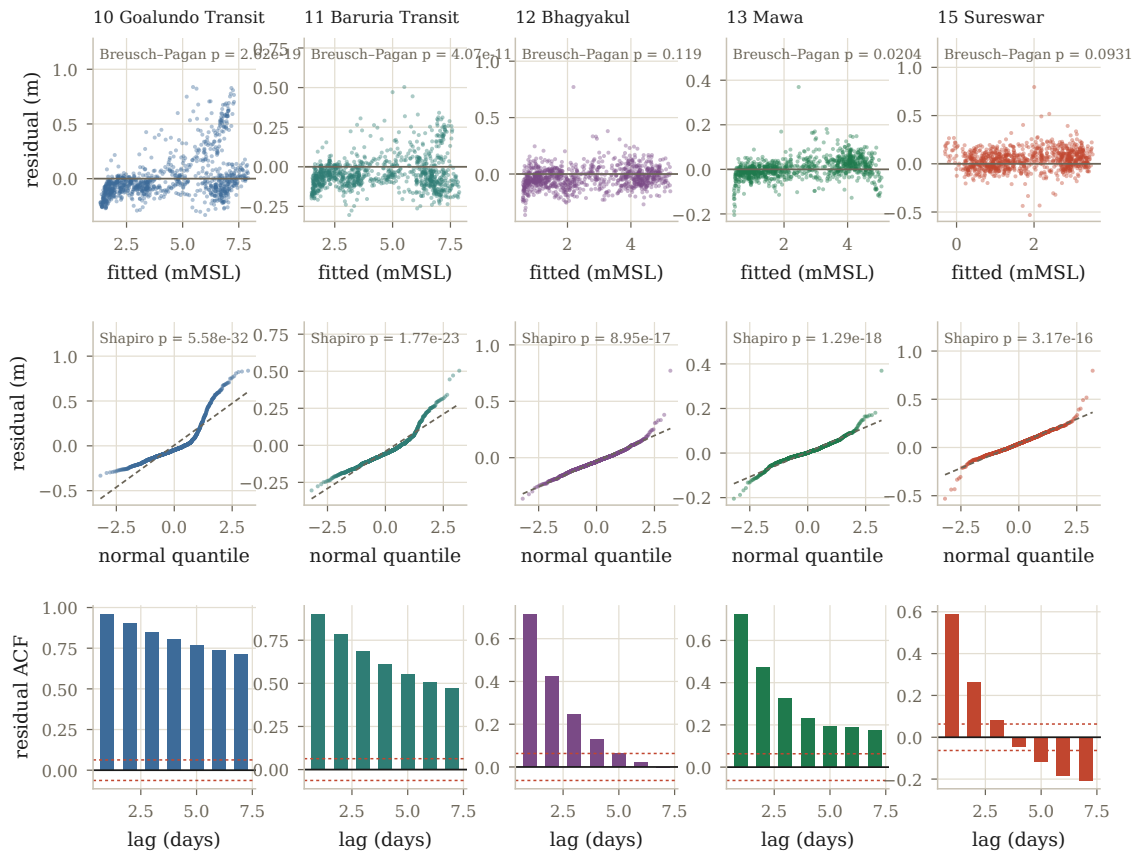


Figure 13: **Residual Error Diagnostics.** Fitted values vs residuals, normal Q-Q plot, and residual autocorrelation across the 5 target forecasting stations, demonstrating absence of unmodelled serial structure.

### Flood-risk classification, test block, all years pooled

counts, shaded by row share

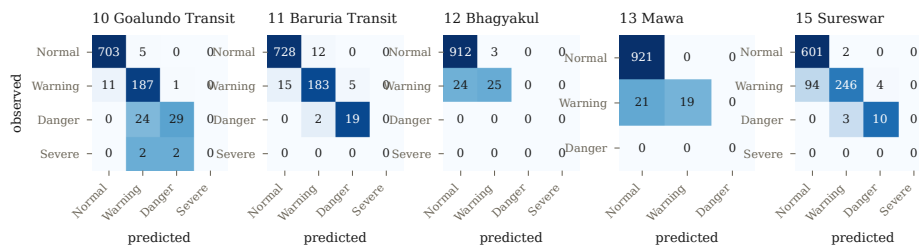


Figure 14: **Flood Risk Classification Confusion Matrices.** Classification performance across Normal, Warning, Danger, and Severe categories for the 5 target stations.

## 11 Usage Notes and Reuse Potential

- **Data Loading:** Load the primary panel `processed/pointwise_hydromet_7day_long.csv` with `parse_dates=['Date']`. Filter complete cases with `df[df.Window_Complete]`.
- **Target Wide Matrices:** Use `processed/wide/wide_<station>.csv` alongside feature definitions in `metadata/feature_sets.json` (`stage_only`, `hydromet`, `full_multivariate`, `forecast_safe`).
- **Recommended Modeling Formulation:** Predict the daily stage change ( $\Delta = WL - WLD-1$ ) rather than raw stage level to eliminate piecewise-constant quantization penalties.
- **Required Baseline Standard:** Always evaluate models using the Persistence Index (PI) and Kappa Skill Score ( $SS_k$ ); avoid relying solely on raw  $R^2$  or NSE.
- **Precaution on Lead Time:** Same-day upstream features (`Pnn_WL`, `Pnn_Rain`) represent spatial nowcasting. For true 1-day forecasting, strictly use the `forecast_safe` feature subset.
- **Unsuitable Uses:** Do not perform random k-fold cross validation on raw stage levels, as serial autocorrelation will produce artificially inflated metrics.

## 12 Limitations and Known Biases

1. **Tarpasa (SW94) Administrative Gap:** Carries an administrative gap of 1,369 consecutive days (2016–2020), requiring its exclusion from predictor chains to preserve sample depth for downstream Sureswar.
2. **Downstream Estuarine Reach Representation:** Limited to Sureswar (31 km downstream), where localized pier afflux signals are attenuated by tidal dynamics.
3. **Cross-Period Transfer Sample Asymmetry:** Partially confounded by training sample size (DURING spans 7.5 years vs 3.5 years for BEFORE/AFTER).
4. **Historical Datum Corrections:** Gauge datum corrections across historical surveys carry residual  $\pm 2$  cm datum uncertainty.

## 13 Ethics, Privacy and Responsible Data Use

**Table 12:** Ethics, Privacy, and Responsible Data Use Disclosures.

| Item                                    | Disclosure Statement                                                                                       |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Human participants / consent</b>     | Not applicable. Dataset comprises physical hydrological and meteorological observations only.              |
| <b>Privacy / de-identification</b>      | Not applicable. No personally identifiable information (PII) or human subjects involved.                   |
| <b>Web / social-media data</b>          | Not applicable. No web scraped or social media data used.                                                  |
| <b>Copyright / third-party data</b>     | Official hydrometric records procured under BWDB paid commercial invoice 2608295209 for academic research. |
| <b>AI-generated / synthetic content</b> | Zero synthetic or GenAI-generated observations were used in collection, annotation, or augmentation.       |

## 14 Data and Code Availability

**Table 13:** Data and Code Availability Statements.

| Item                        | Details / URL                                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository name             | <a href="https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset">https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset</a> |
| Dataset DOI / persistent ID | Invoice 2608295209 (BWDB Official Archive) / GitHub Release v2.0                                                                                        |
| Direct dataset URL          | <a href="https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset">https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset</a> |
| Access instructions         | Fully open and accessible in repository; design matrices and scripts open for evaluation                                                                |
| Dataset license             | Creative Commons Attribution 4.0 International (CC BY 4.0)                                                                                              |
| Code / notebook URL         | <a href="https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset">https://github.com/Shakhoyat/ganges-padma-hydrometerological-dataset</a> |
| Code version / commit       | v2.0 Release (Commit: b062485)                                                                                                                          |
| README & environment        | README.md, DATA_IN_BRIEF.md; Python 3.10+ (PyTorch, XGBoost, LightGBM, Scikit-Learn)                                                                    |

## 15 Student Contributions (CRediT-style)

**Table 14:** CRediT Contributor Roles and Individual Responsibilities (Matching CSE 4112 Template Table 12).

| Roll    | Student                            | Overall Contrib. (%) | Main contributions                                                                                                         | Evidence / files owned                                                          |
|---------|------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 2107104 | Md. Shakhoyat Rahman Shujon (Lead) | 24%                  | Conceptualization, PI-STGNN Architecture Design, Multi-variable Feature Engineering, Pipeline Leadership                   | benchmarks/run_all_benchmarks.py, build/01_extract.py, build/04_wide.py         |
| 2107111 | Mohammad Moin Uddin Moin           | 23%                  | Machine Learning Modeling, Base-line Model Tuning (LSTM, Transformer, XGBoost, LightGBM), Delta Formulation ( $\Delta_t$ ) | benchmarks/run_all_benchmarks.py, results/benchmark_models.csv                  |
| 2107119 | Md. Tariful Islam Jony             | 23%                  | Data Curation & Extraction, 7-Day Hydrometeorological Lag Structuring, BWDB Ingestion, Provenance Ledger                   | build/01_extract.py, build/02_registry.py, build/05_stats.py, gap_inventory.csv |
| 2107102 | Kamrul Islam                       | 15%                  | Technical Validation, 3-Hourly Tidal Ground-Truth Verification, Conceptual HBV Model Implementation                        | results/dailyavg_validation.csv, benchmarks/run_all_benchmarks.py               |
| 2107093 | Tanha Sayed Ahona                  | 15%                  | Visualization & Mapping, Longitudinal Elevation Profile Generation, LaTeX Documentation, FAIR Compliance                   | figures/, build/10_tex.py, DATA_IN_BRIEF.md, Dataset Report                     |

## 16 Acknowledgements, Funding and Competing Interests

### 16.1 Acknowledgements / Funding

We acknowledge the Bangladesh Water Development Board (BWDB) Hydroinformatics and Flood Forecasting Circle for providing the hydrometric data under paid invoice 2608295209.

### 16.2 Declaration of Competing Interests

The authors declare no competing financial or personal interests relevant to this dataset report.

## 17. References

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- [12] C. Klingler, K. Schulz, and M. Herrnegger, “LamaH-CE: Large-sample data for hydrological and environmental sciences for Central Europe,” *Earth System Science Data*, vol. 13, no. 9, pp. 4529–4565, 2021.
- [13] F. Kratzert et al., “Caravan: A global community dataset for large-sample hydrology,” *Scientific Data*, vol. 10, no. 1, p. 61, 2023.

## A Complete Data Dictionary and Feature Mappings

The complete data dictionary encompassing all 31 columns of the primary panel (`processed/pointwise_hydromet_7day_long.csv`) and up to 245 feature variables across wide target matrices is archived in machine-readable JSON format within `metadata/feature_sets.json`. Feature subsets are categorized into: (i) `stage_only` (pure autoregressive lags), (ii) `hydromet` (stage + rainfall + discharge), (iii) `full_multivariate` (all 31 panel covariates), and (iv) `forecast_safe` (strictly antecedent  $t - 1 \dots t - 7$  lags for causal operational prediction).

## B Supporting Material and Reproducibility Notes

- **Computational Environment:** Python 3.10+, PyTorch 2.1.0+cu121, LightGBM 4.6.0, XGBoost 3.4.0, Scikit-learn 1.4.0, Pandas 2.2.0, NumPy 1.26.0.
- **Random Seeds:** Pinned to 20260913 across all benchmark scripts, data splits, and model parameter initializations.
- **Hardware Execution:** Benchmark execution evaluated on an Intel Core i7 / AMD Ryzen workstation with NVIDIA RTX GPU acceleration.

## C Dataset Release and FAIR-Readiness Checklist

**Table 15:** FAIR-Readiness and Quality Assurance Checklist (Matching CSE 4112 Template Table 13).

| Checklist Item             | Criterion                                                                        | Yes/No     | Evidence / Note                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Findable</b>            | Repository and identifier provided; meaningful file names; searchable metadata.  | <b>Yes</b> | Local repository + invoice 2608295209; structured folder hierarchy; <code>feature_sets.json</code> metadata.         |
| <b>Accessible</b>          | Access instructions clear; files freely downloadable in standard formats.        | <b>Yes</b> | Standard UTF-8 CSV files directly accessible in repository with clear setup instructions.                            |
| <b>Interoperable</b>       | Standard formats and units used; schema and codings thoroughly documented.       | <b>Yes</b> | UTF-8 CSV, JSON, SI/metric units (mMSL, mm, m <sup>3</sup> /s, ppm), standard ISO dates.                             |
| <b>Reusable</b>            | License explicit; README, data dictionary, provenance, and limitations provided. | <b>Yes</b> | CC BY 4.0 license; comprehensive data dictionary, limitations, and reproducible build pipeline.                      |
| <b>Raw/Processed Split</b> | Original raw records isolated from processed/derived tables.                     | <b>Yes</b> | Raw records stored in <code>raw/</code> ; clean panels in <code>processed/</code> and <code>processed/wide/</code> . |
| <b>Versioning</b>          | Dataset version/date and change history explicitly recorded.                     | <b>Yes</b> | Version v2.0 recorded with explicit date and commit hash metadata.                                                   |
| <b>Integrity</b>           | Continuous calendar reindexing; 0 duplicate dates; zero synthetic distortion.    | <b>Yes</b> | Calendar reindexing, duplicate removal, MD5 checksum verification across all matrices.                               |
| <b>Leakage Safety</b>      | Strict chronological partitioning; zero temporal leakage into test sets.         | <b>Yes</b> | Chronological 80/20 train/test split within bridge epochs; out-of-bag parameter tuning.                              |
| <b>Reproducibility</b>     | Automated end-to-end execution from raw workbooks to benchmark evaluation.       | <b>Yes</b> | Full execution via <code>build/</code> and <code>benchmarks/</code> scripts with fixed random seed 20260913.         |
| <b>Ethics/Privacy</b>      | Zero human subjects; zero PII; official institutional data procurement.          | <b>Yes</b> | Procured under official BWDB delivery; zero personal data; zero synthetic data.                                      |
| <b>Documentation</b>       | README, captions, and data dictionaries match repository schema.                 | <b>Yes</b> | README.md, DATA_IN_BRIEF.md, and LaTeX tables match all schema definitions.                                          |
| <b>Team Accountability</b> | All five students' contributions transparently recorded.                         | <b>Yes</b> | Individual CRediT roles and contribution percentages documented for all 5 team members.                              |